

What we set out to do

Our research question asks whether retrieval-augmented generation (RAG) can produce better Brief Hospital Course summaries than naive zero-shot prompting when working with fragmented, multi-source ICU clinical notes. We used 100 ICU patients from the VeriFact-BHC dataset (MIMIC-III), each with dozens of clinical notes across 14 categories — nursing, physician, radiology, respiratory, ECG, and others. Each patient also has a human-written BHC that serves as our gold standard, plus 13,070 clinician-annotated propositions rating each factual claim as Supported, Not Supported, or Not Addressed.

The core clinical motivation is straightforward: these patients have a median of 25 notes totaling ~17,500 tokens, with the most complex patients generating over 200,000 tokens of documentation. No human reads all of that. A good BHC requires synthesizing the key events across all of it — and that's what we're testing whether an LLM can do.

What we built

The baseline is the simplest possible approach. We concatenate all notes in chronological order and hand them to Mistral 7B Instruct (v0.3, 4-bit quantized, 32K context window) with a one-line instruction: "write a Brief Hospital Course." No retrieval, no examples, no structure. For 42 of our 100 patients, the notes exceeded 32K tokens and the model only saw the beginning — everything after got cut off. This is the approach we're trying to beat.

The RAG pipeline replaces blind concatenation with intelligent selection. We chunk all 4,787 notes into 23,663 pieces (~300 tokens each), embed them with BGE-M3 (a biomedical embedding model), and build a per-patient FAISS index for fast similarity search. At generation time, we query each patient's index with six clinically-motivated queries — one each for admission, diagnoses, workup, treatment, clinical course, and discharge — retrieve the most relevant chunks, sort them chronologically, and feed them to the same Mistral 7B with the same generation settings. The only variable is what content goes in.

We iterated through three RAG versions. The first used a single generic query and retrieved mostly medication lists and ICU order templates — semantically similar to "hospital course" but not the narrative content a BHC needs. The second switched to six targeted queries mirroring how physicians actually organize a hospital course, which improved content diversity. The third matched the token budget to what baseline gets (~28K tokens), after we realized our initial RAG was feeding the model one-third the context baseline had — an unfair comparison.

What we found

On automated metrics, RAG and baseline are essentially equivalent. ROUGE-L: 0.168 (RAG) vs 0.165 (baseline). BERTScore: 0.820 vs 0.819. These differences are within noise. RAG did not meaningfully improve lexical or semantic overlap with the human-written BHC.

The subgroup analysis we expected to be decisive wasn't. We hypothesized that RAG would help most for the 42 patients who were truncated in baseline — the ones losing data. It didn't. Truncated patients showed 19/42 improved and 20/42 worsened with RAG. Not truncated patients: 19/58 improved, 32/58 worsened. No clear pattern.

Proposition-level fact recall told the same story. We used the 5,396 clinician-verified "Supported" propositions from the VeriFact dataset as a factual checklist. For each patient, we measured what fraction of these verified facts appeared in the generated BHC using semantic similarity matching. Baseline captured 73.5% of facts; RAG captured 72.5%. No meaningful difference.

But qualitative review revealed something the metrics missed. When we read individual outputs side by side, RAG reduced hallucinations that ROUGE cannot detect. The clearest example: Patient 72940, a CABG patient with 37K tokens of notes. Baseline hallucinated that the patient was "admitted for aortoiliac disease" — a fabricated diagnosis. RAG correctly identified the CABG with specific graft details (LIMA-LAD, SVG-OM) that match the human-written BHC word for word. ROUGE-L for this patient jumped from 0.113 (baseline) to 0.240 (RAG). Conversely, for Patient 61157 (subarachnoid hemorrhage, 86K tokens), baseline did better — it captured the narrative arc from the early notes while RAG produced a longer but vaguer summary.

Why RAG didn't beat baseline on ROUGE

Three factors explain this:

First, chronological truncation is a surprisingly effective strategy. When baseline truncates, it keeps the earliest notes — admission, ED evaluation, initial workup, early assessments. These contain the chief complaint, diagnoses, and clinical reasoning that form the backbone of any BHC. The notes that get cut are later in the stay: daily progress notes that are often repetitive, routine nursing assessments, and repeat vital sign documentation. Losing these hurts less than we expected.

Second, RAG disrupts temporal coherence. A BHC is a narrative — "the patient came in with X, we found Y, we treated with Z, they improved/worsened, and were discharged in this condition." Baseline preserves this chronological flow because the notes are in order. RAG selects chunks from across the stay based on relevance, and even though we re-sort them by date, the narrative continuity between consecutive chunks is broken. The model has to reconstruct a story from fragments rather than reading a continuous record.

Third, ROUGE measures word overlap, not factual accuracy. A generated BHC that confidently states the wrong diagnosis will score similarly to one that gets it right, as long as both use similar medical vocabulary. The hallucination in Patient 72940 — "aortoiliac disease" instead of CABG — doesn't tank the ROUGE score because the surrounding text still uses words like "surgery," "postoperative," and "recovery" that overlap with the reference.

How this aligns with the literature

Our finding is not an anomaly. Kruse et al. (2025, EMNLP Findings) ran almost the same experiment — Mistral 7B with and without RAG on MIMIC-III discharge summarization — and found RAG *decreased* Mistral's ROUGE-L by 1.24 points while *increasing* Llama3's by 4.08 points. The RAG effect is model-dependent, not universally beneficial. Importantly, when they evaluated with PDSQI-9 (a clinician-validated 9-dimension quality rubric) using LLM-as-Judge, RAG outperformed direct generation on accuracy, thoroughness, and usefulness — dimensions ROUGE cannot capture.

Alkhalaf et al. (2024, Journal of Biomedical Informatics) showed RAG improving summarization accuracy from 93.25% to 99.25% — but they measured this through manual clinical verification, not ROUGE. They didn't even report ROUGE scores. Their accuracy improvement would have been invisible to automated metrics.

The VeriFact paper itself (Chung et al. 2025, NEJM AI) demonstrated that LLM-generated BHCs achieve 95% factual support from the EHR while human-written BHCs only achieve 62.6% — because clinicians incorporate information from bedside conversations, phone calls, and other sources not in the chart. This creates a natural ceiling on how well any model can match the human BHC using only EHR data.

Multiple groups are converging on the same conclusion: ROUGE and BERTScore are necessary but insufficient for evaluating clinical text generation. The field is moving toward proposition-level verification and multidimensional clinical quality instruments.

What this means for Phase B

Our Phase A results directly motivate three things for the final project:

Better evaluation is the priority. We have the VeriFact proposition annotations and should implement PDSQI-9 / LLM-as-Judge evaluation. Kruse et al. showed this is where RAG's benefit becomes visible. If we can demonstrate that RAG improves clinical quality on dimensions ROUGE misses, that's a strong paper-worthy finding.

Multi-model comparison would strengthen the story. Kruse et al.'s model-dependent finding (RAG helps Llama3 but hurts Mistral) suggests testing with Llama3 could reveal an improvement our current setup masks. This is planned pending HuggingFace access.

Structured output templates may amplify RAG's benefit. Kruse et al. found that structured, focused tasks (Assessment & Plan generation) benefited more from RAG than broad discharge summaries. Our Phase B case-presentation template — which constrains output into admission, workup, treatment, course, and discharge sections — aligns with this. Generating section by section with section-specific retrieval may be where RAG really shines.

Quick reference numbers

	Baseline	RAG v3
ROUGE-L F1	0.165	0.168
BERTScore F1	0.819	0.820
Fact recall	73.5%	72.5%
Context used	~31K tokens	~18K tokens
Patients truncated	42/100	0/100
Runtime (A100)	65.6 min	62.8 min
Model	Mistral 7B v0.3, 4-bit	Same
Dataset	VeriFact-BHC, 100 patients	Same